DGRF 2015 Evaluation

I use here different criteria described in the 2015 paper about the IGRF/DGRF evaluation. At the moment, I performed a gross analysis only on the DGRF as I must be careful reformatting the initial candidate models. The two other analyses will follow.

Principles: The candidate models for epoch 2015.0 are compared to the arithmetic mean model. Other comparisons can be made at a later stage: either by considering the median model or by using first the Huber weighting scheme in space used for IGRF-12.

The figure in the geographical domain are sketched at the Earth's mean radius.

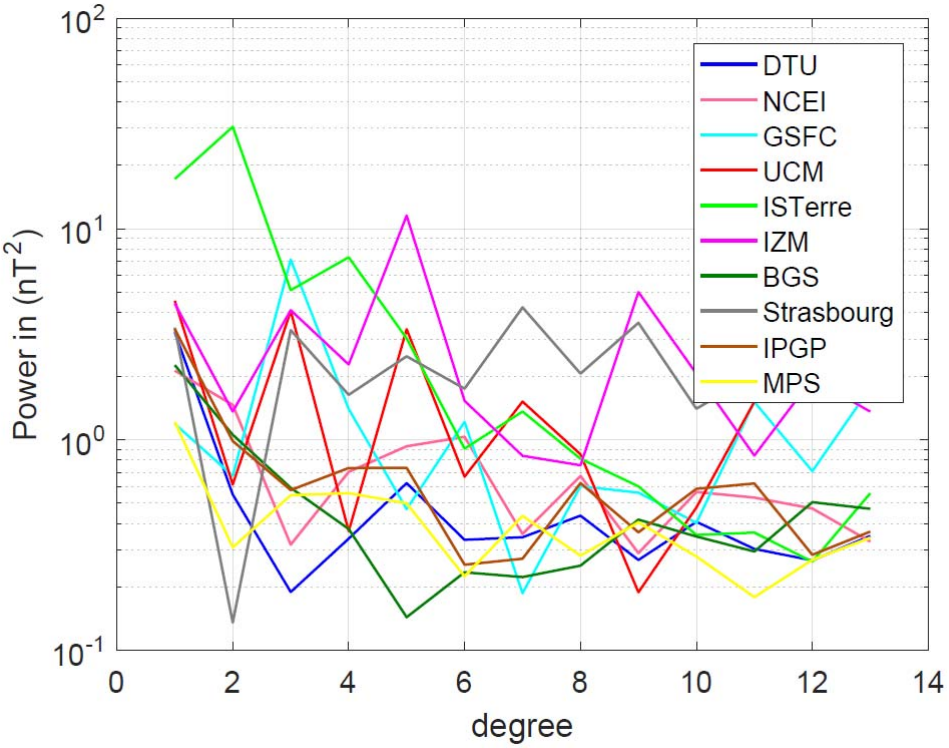
The models are analyzed in the following order:

1:DTU,
2:NCEI
3:GSFC
4:UCM
5:ISTerre
6:IZM
7:BGS
8:Strasbourg
9:IPGP
10:MPS
11:GFZ

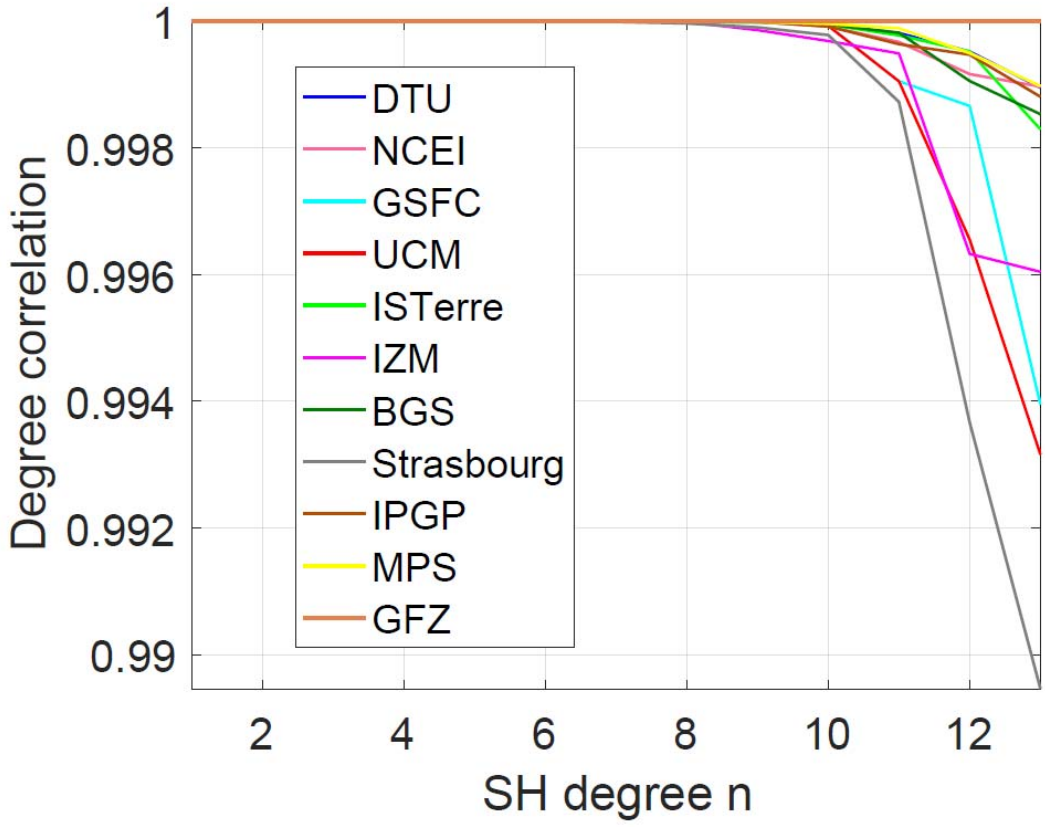
The last column is the arithmetic mean model. Attached is the matrix of RMS between all models corresponding to Ciaran's plot (in a different order).

Summary: There are two groups of models. The average RMS is close to 4 nT. Three out of the 11 candidate models have RMS significantly larger between 6-9 nT. These models are ISTerre, IZMIRAN, STRASBOURG. The spectral analyses indicate that the difference arise mostly for degrees 1 to 2 for ISTerre, for zonal terms (m=0) for IZMIRAN, and on all coefficients for STRABOURG. The differences in the geographical domain show that the GSFC model has structures following the magnetic equator. This probably comes from the parameterization including the induced internal field coefficients. For IZMIRAN it seems clear that the external ionospheric field was not corrected for correctly as the structures look like the Sq signal. For ISTerre, the residuals are more peculiar over over the South Atlantic with a negative structures that I do not fully understand but also over the Southern cap that may come from a magnetospheric field contamination. The residuals for the STRASBOURG model shows differences more prominent at smaller scales by downward continuation to the Earth's surface.

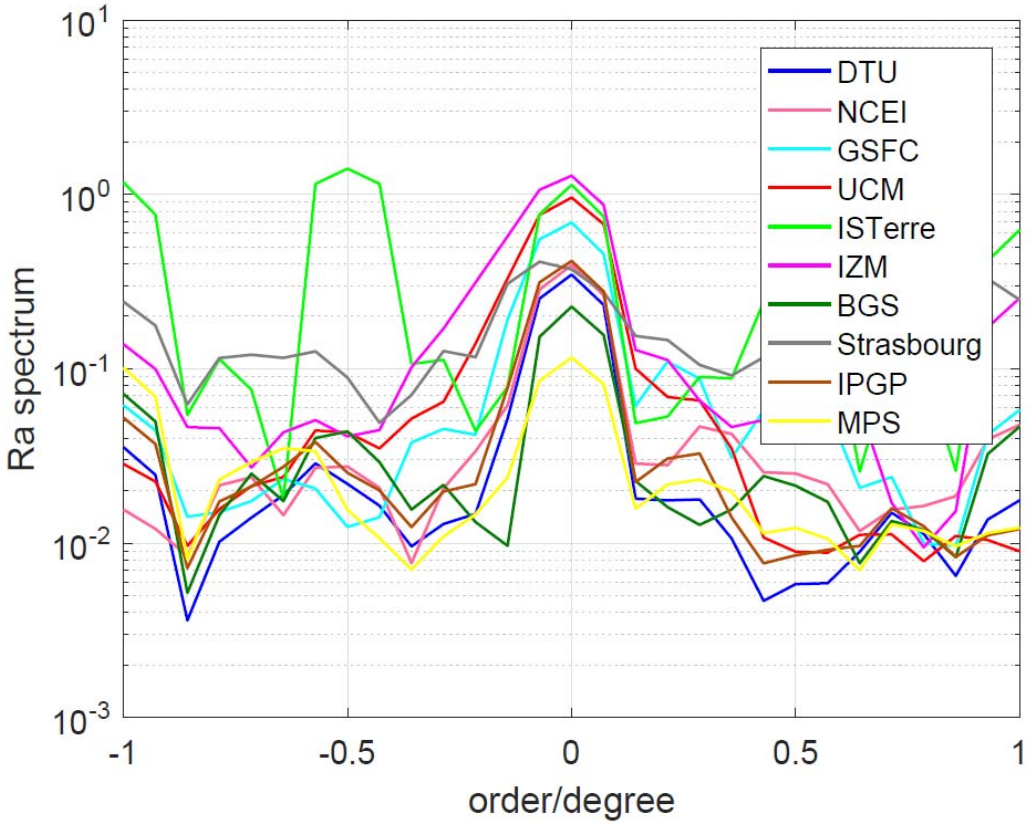
The power spectra of the difference to the mean are:



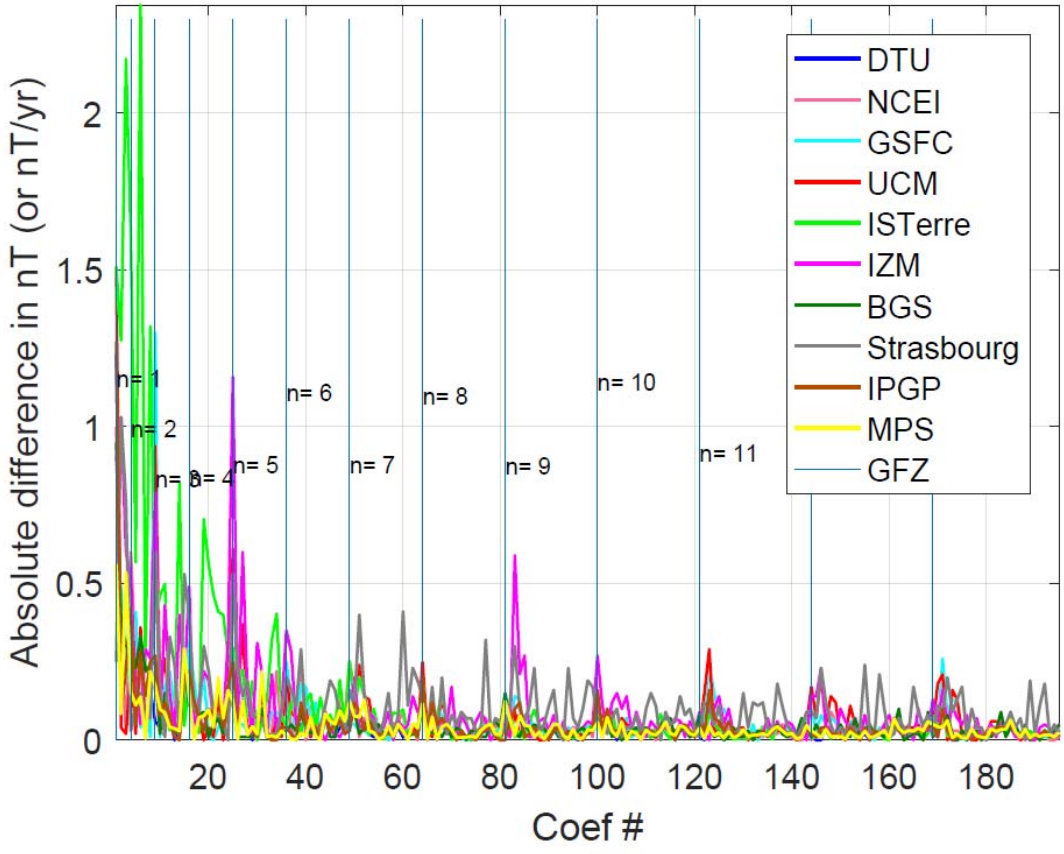
The correlation between each candidate model to the mean is:



The azimuthal power spectrum is:

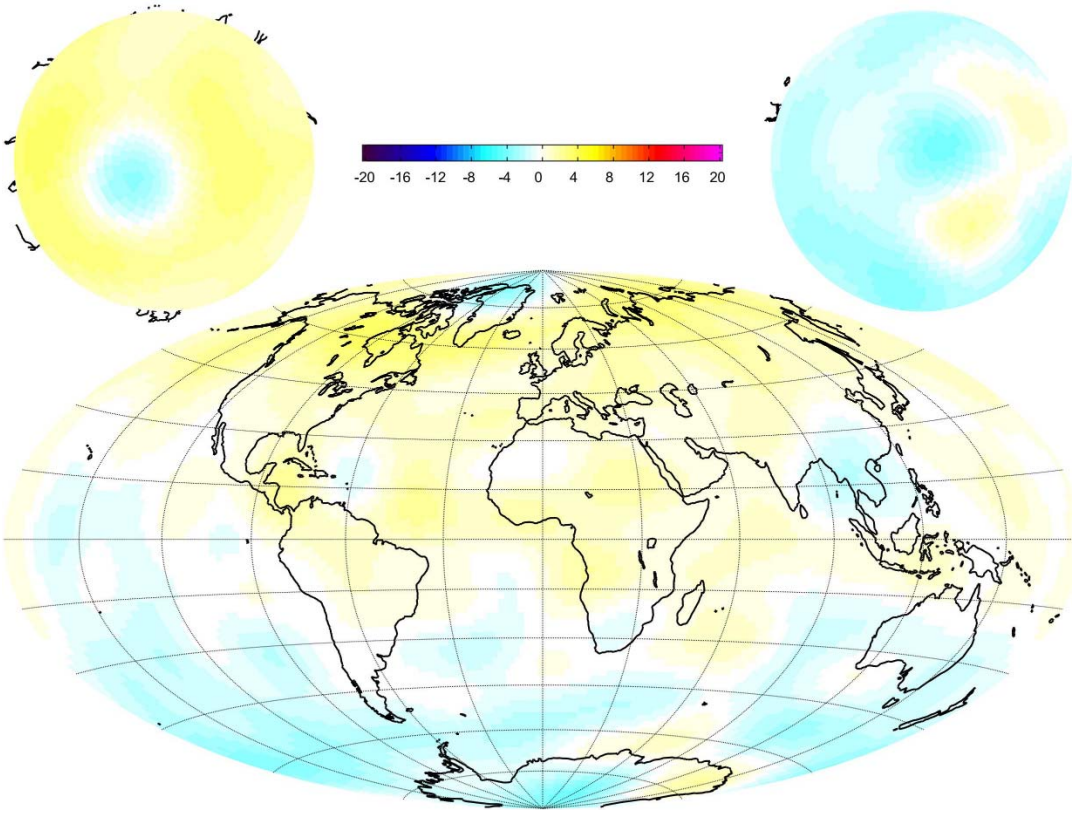


The coefficient difference is:

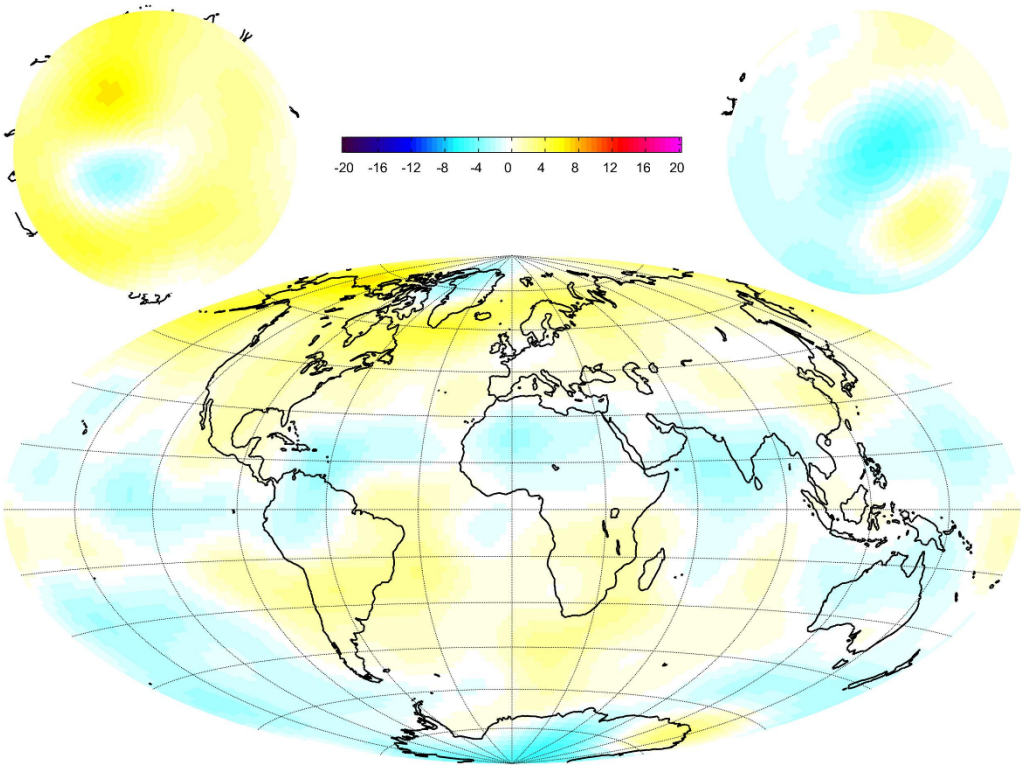


And the differences in the geographical domain for the radial component of each candidate model (at the Earth's mean radius) is:

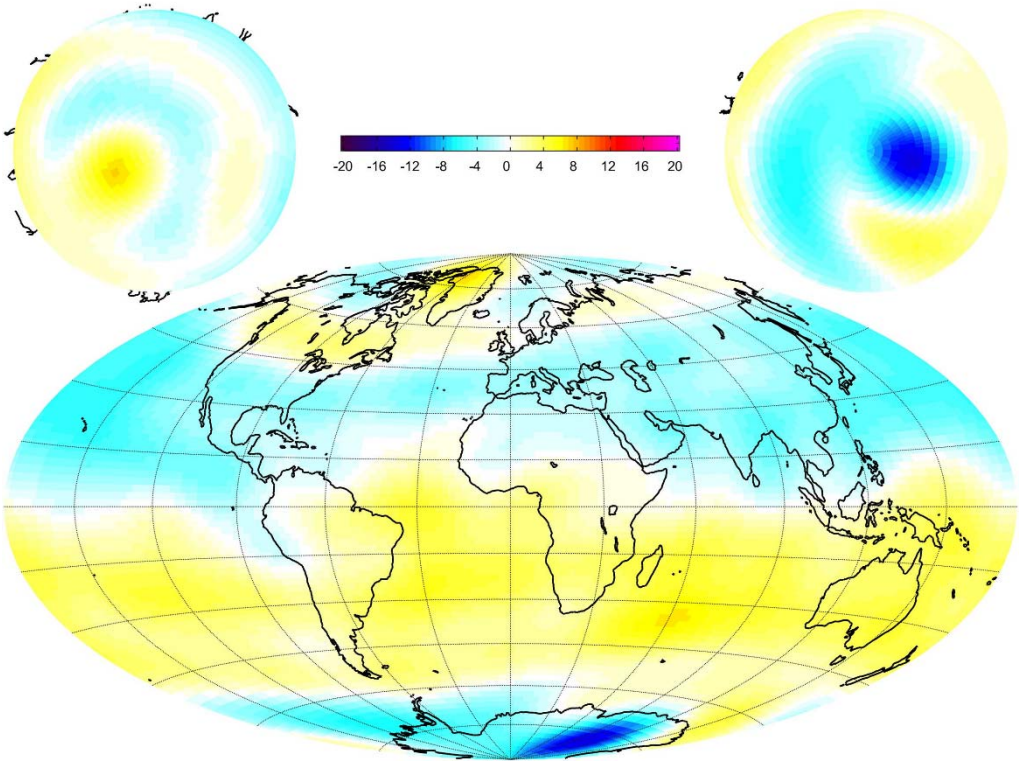
DTU:



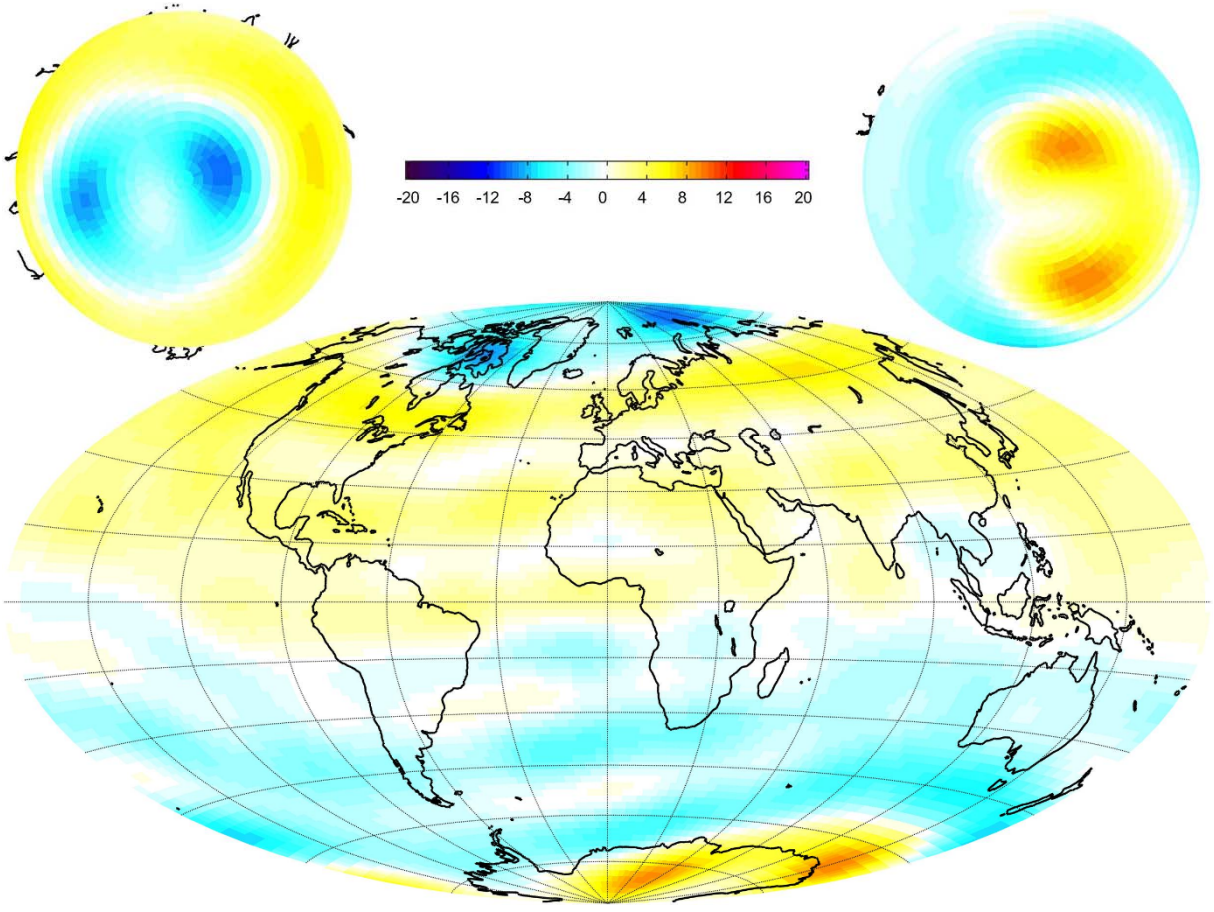
NCEI:



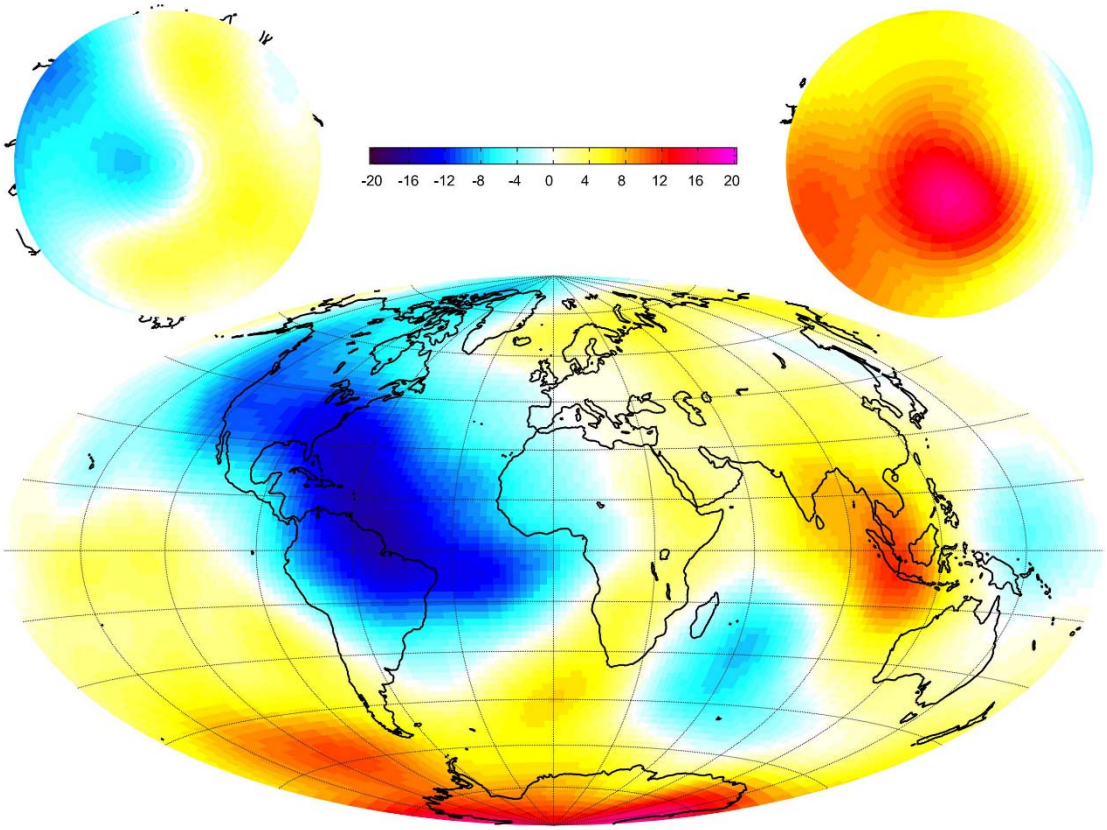
GSFC:



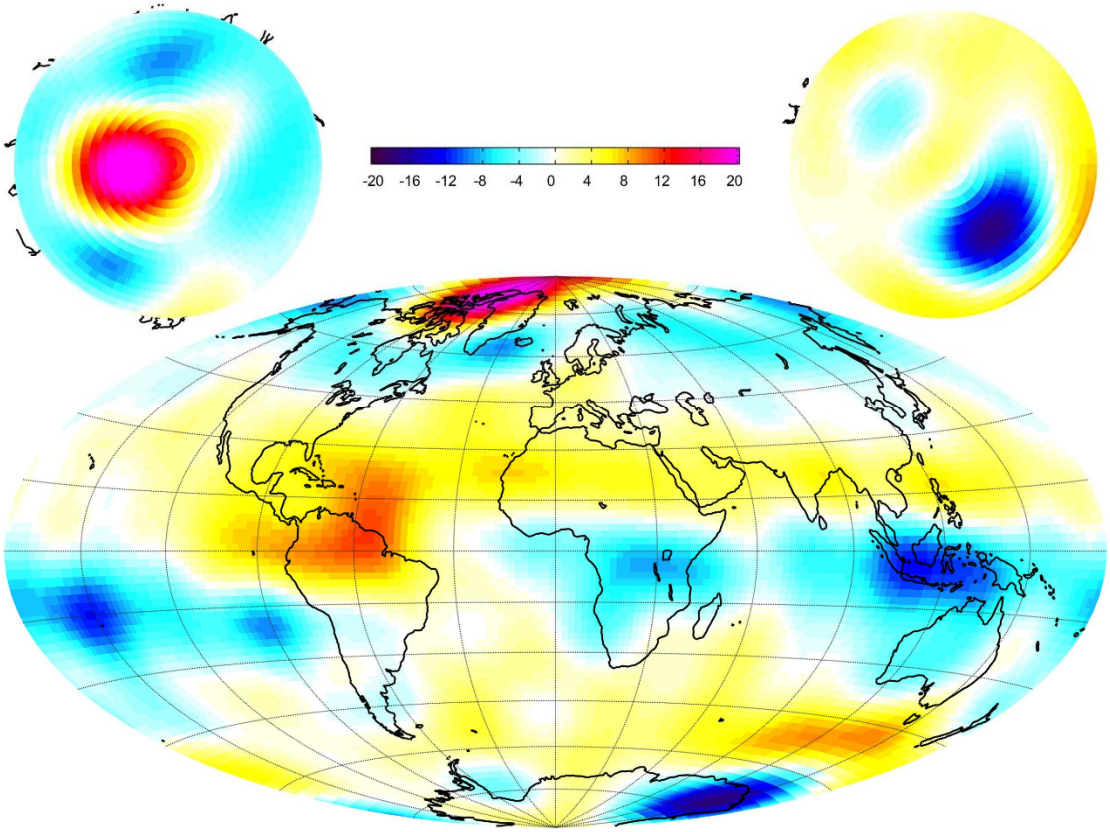
UCM:



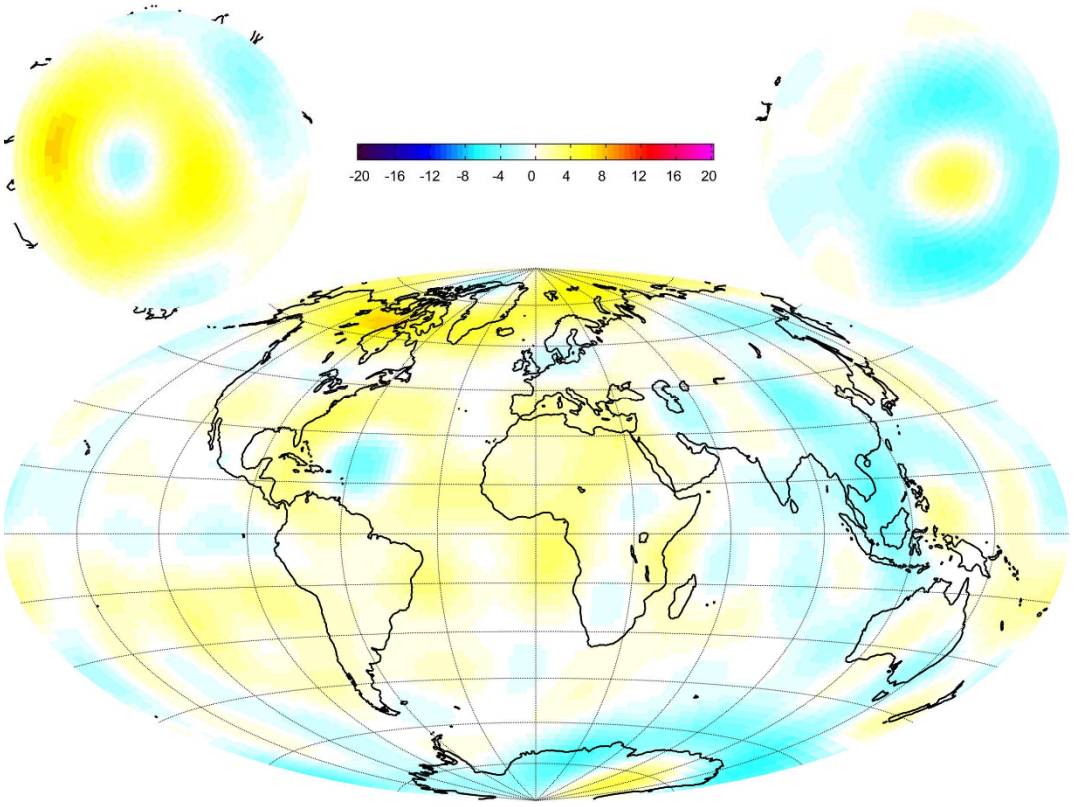
ISTerre:



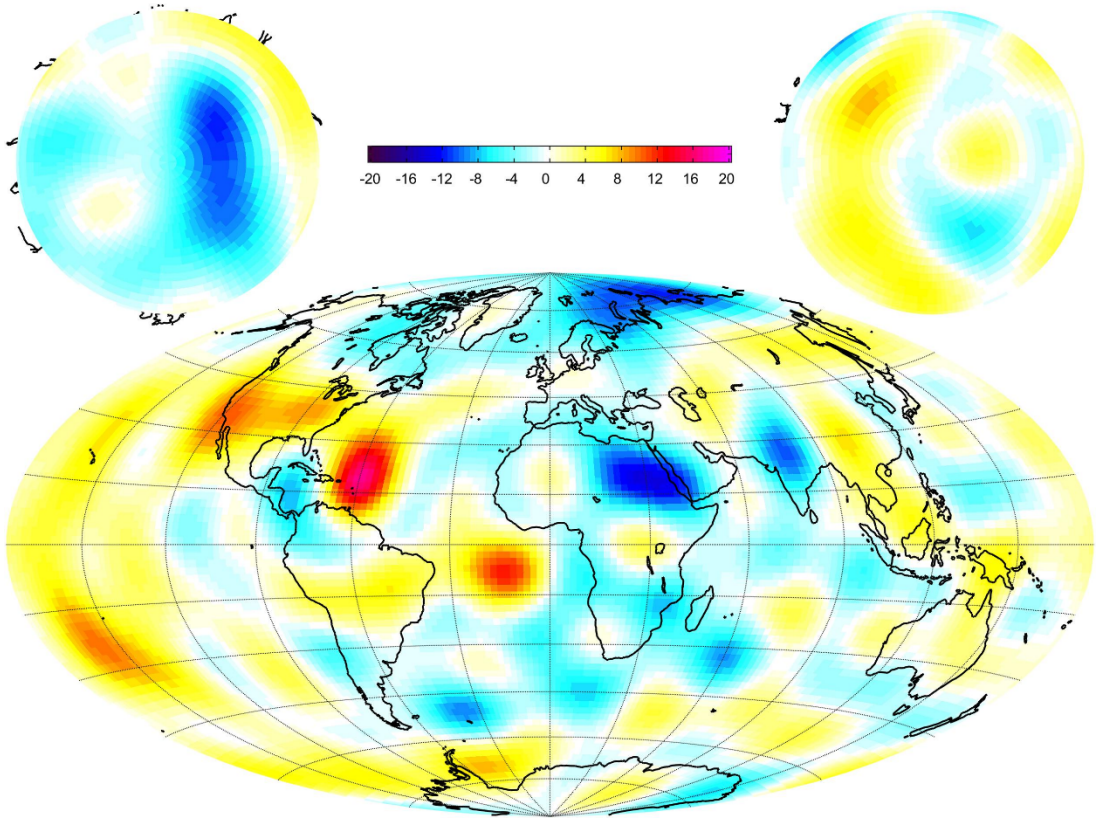
IZMIRAN:



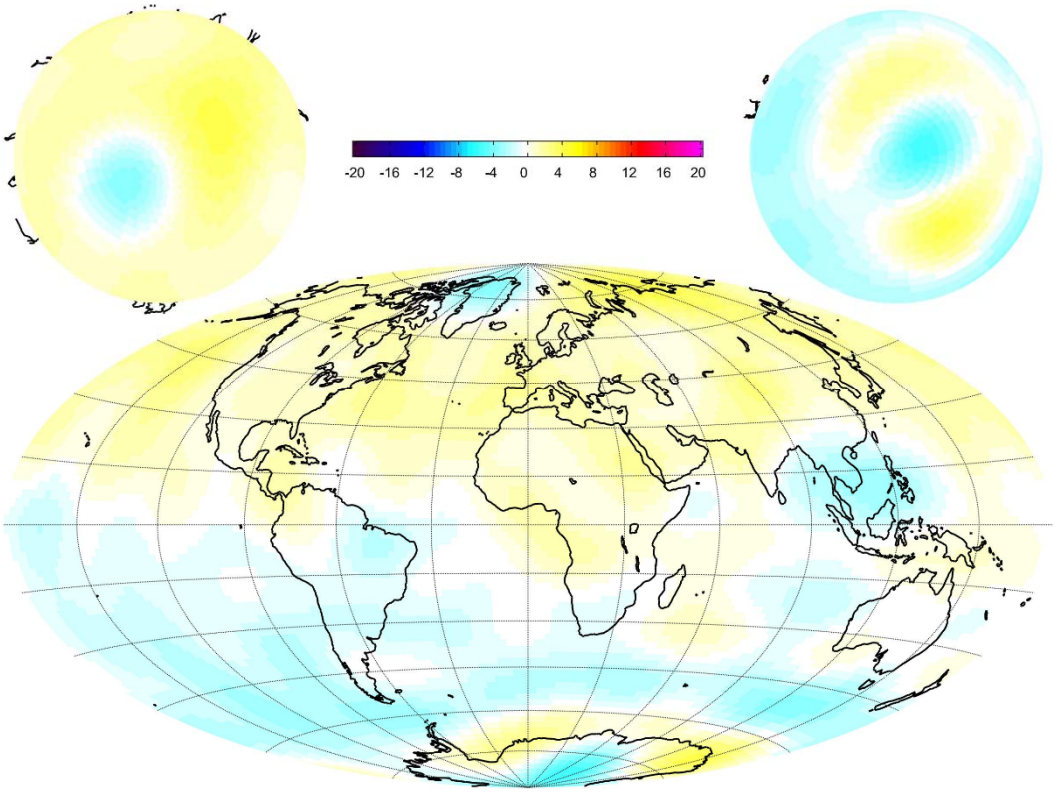
BGS:



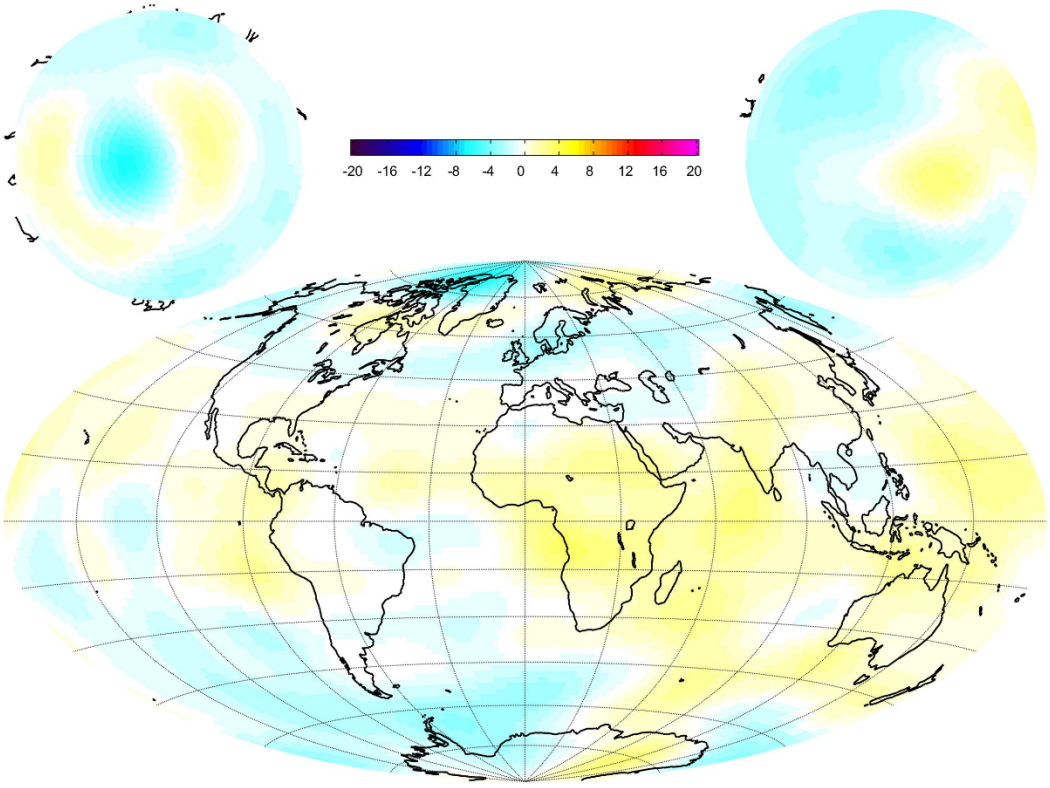
STRASBOURG:



IPGP :



MPS :



GFZ :

